Luca Denti

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Research Interests

My research activity focuses on developing algorithms, data structures, and tools for processing and analyzing WGS data. My interests are in the field of pangenomics, genomic variations analysis, and transcriptome analysis.

Current Position

Assistant professor Sep. 2026 – Present
Department of Applied Informatics, Comenius University *Bratislava, Slovakia*

Past Positions

Postdoctoral Researcher (ERA Fellowship) Sep. 2024 – Aug. 2026
Department of Applied Informatics, Comenius University *Bratislava, Slovakia*
- Supervisor: Prof. Tomas Vinar

Postdoctoral Researcher Mar. 2022 – Aug. 2024
Department of Informatics, Systems and Communication, University of Milano-Bicocca *Milan, Italy*
- Supervisor: Prof. Paola Bonizzoni

Postdoctoral Researcher Sep. 2020 – Feb. 2022
Department of Computational Biology, Institut Pasteur *Paris, France*
- Supervisor: Dr. Rayan Chikhi

Research Fellowship Jan. 2020 – Aug. 2020
Department of Informatics, Systems and Communication, University of Milano-Bicocca *Milan, Italy*
- Supervisor: Dr. Yuri Pirola

Education

Doctor of Philosophy in Computer Science Nov. 2016 – Oct. 2019
University of Milano - Bicocca *Milan, Italy*
- Thesis: "Algorithms for analyzing genetic variability from Next-Generation Sequencing data"
- Supervisors: Prof. Paola Bonizzoni, Dr. Raffaella Rizzi, Dr. Marco Previtali

Master's Degree in Computer Science Nov. 2014 – Oct. 2016
University of Milano - Bicocca *Milan, Italy*
- Thesis: "A new algorithmic approach for the approximate alignment of a string to a graph"
- Supervisors: Prof. Paola Bonizzoni, Dr. Raffaella Rizzi

Bachelor's Degree in Computer Science Oct. 2011 – Oct. 2014
University of Milano - Bicocca *Milan, Italy*
- Thesis: "Development of an enhanced Petri Net simulator"
- Supervisors: Dr. Luca Bernardinello, Prof. Lucia Pomello

Awards

Young Talents Award 2022
University of Milano-Bicocca *Milan, Italy*

Best Poster Award 2018
RECOMB, Research in Computational Molecular Biology *Paris, France*

Research projects

ASVA-CGR: Algorithms for Structural Variation Analysis in Challenging Genomic Regions
Funding: EU Horizon, Widening participation and spreading excellence. Role: Principal investigator

PANGAIA: Pan-genome Graph Algorithms and Data Integration
Funding: EU Horizon 2020, RISE Marie Skłodowska-Curie Action. Role: administrative and research

Visiting activities

- **Visiting Experienced Researcher** at Geneton S.r.o, Bratislava (Slovakia) - PANGAIA project
  Efficient algorithm for computational comparative pangenomic with Prof. Tomas Szemes [Aug. 2023 / Jan. 2024]
- **Visiting Experienced Researcher** at University of Tokyo, Tokyo (Japan) - PANGAIA project
  Efficient algorithm for computational comparative pangenomic with Prof. Kunihiko Sadakane [Feb. 2023 - June 2024]
- **Visiting Experienced Researcher** at Geneton S.r.o, Bratislava (Slovakia) - PANGAIA project
  Efficient algorithm for computational comparative pangenomic with Prof. Tomas Szemes [Sept/Dec 2022]
- **Visiting Ph.D. student** at Centrum Wiskunde & Informatica, Amsterdam (The Netherlands)
  Online pattern matching in elastic degenerate strings with Prof. Leen Stougie and Dr. Solon Pissis [Feb/Apr 2019]
- **Visiting Ph.D. student** at Centrum Wiskunde & Informatica, Amsterdam (The Netherlands)
  Mapping-free genotyping of short variations with Prof. Alexander Schönhuth [Oct/Dec 2018]

Talks and Posters

Invited Talks:

4. Pangenome graph augmentation from unassembled long reads. Jan. 23, 2025. Pangenome Bio Hacking 2025 Workshop - Online
3. Structural Variations Discovery from Specific Strings. Feb. 15, 2024. IRISA / Inria RBA, Rennes - France (invited by: Pierre Peterlongo)
2. Structural Variations Discovery from Specific Strings. Nov. 9, 2022. Comenius University, Bratislava - Slovakia (invited by: Tomas Vinar)
1. Genotyping by mapping-free allele detection of known variants. Dec. 15, 2019. Institut Pasteur, Paris - France (invited by: Rayan Chikhi)

Talks:

11. Pangenome graph augmentation from unassembled long reads
    Information Technologies - Applications and Theory (2025). Telgárt, Slovakia
10. Pangenome graph augmentation from unassembled long reads
    Research in Computational Molecular Biology (2025). Seoul, South Korea
9. ESGq: Alternative Splicing Events Quantification across Conditions based on Event Splicing Graphs
   Information Technologies - Applications and Theory (2023). Tatranské Matliare, Slovakia
8. Structural variation discovery in hard-to-call genomic regions using sample-specific strings from accurate long reads
   Research in Computational Molecular Biology (2023). Istanbul, Turkey
7. Comparative genome analysis using sample-specific string detection in accurate long reads
   High Throughput Sequencing Algorithms and Applications (2021). Online
6. Comparative genome analysis using sample-specific string detection in accurate long reads
   Data Structures in Bioinformatics (2021). Online
5. Dynamic quasi-minimal perfect hash function for k-mers
   Data Structures in Bioinformatics (2020). Rennes, France
4. MALVA: genotyping by Mapping-free ALlele detection of known VAriants
   Next Generation Sequencing - Milano (2019). Milano, Italy
3. MALVA: genotyping by Mapping-free ALlele detection of known VAriants
   Annual Meeting of the Bioinformatics Italian Society (2019). Palermo, Italy
2. MALVA: genotyping by Mapping-free ALlele detection of known VAriants
   RECOMB Satellite Workshop on Massively Parallel Sequencing (2019). Washington DC, USA
1. Mapping RNA-seq data to a transcript graph via approximate pattern matching to a hypertext
   Algorithms for Computational Biology (2017). Aveiro, Portugal

Posters:

5. Challenges and Pitfalls in the Evaluation of Structural Variants Callers
   RECOMB and RECOMBSEQ (2026). Thessaloniki, Greece
4. Pangenome-based characterization of novel genetic variants
   MatFyz Connections (2025). Comenius University in Bratislava, Slovakia

3. Alignment-free de novo structural variations from pangenomes
   MatFyz Connections (2024). Comenius University in Bratislava, Slovakia
2. MALVA: genotyping by Mapping-free ALlele detection of known VAriants
   RECOMB (2019). Washington DC, USA
1. ASGAL: aligning RNA-Seq data to a splicing graph to detect novel alternative splicing events
   RECOMB (2018). Paris, France

Teaching

Teacher

Comenius University — *Bratislava, Slovakia*

- Data Management, Comenius University (a.y. 2025/26)
- Methods in Bioinformatics, Comenius University (a.y. 2026/27, 2025/26)
- Seminars in Bioinformatics, Comenius University (a.y. 2026/27)

University of Milano-Bicocca — *Milan, Italy*

- Genome Assembly, "qOmics: quantitative methods for Omics Data" 2024 (Master/Post-graduate course)
- C - Laboratory, *B.Sc. in Physics*, a.y. 2023/24, 2022/23
- Algorithms and Data Structure - Laboratory, *B.Sc. in Computer Science*, a.y. 2021/22

Teaching assistant

University of Milano-Bicocca — *Milan, Italy*

- Programming Language Laboratory, *B.Sc. in Statistics* (a.y. 2019/20)
- Foundations of Computer Science, *M.S. in Data Science* (a.y. 2019/20)
- Design and Analysis of Algorithms, *B.Sc. in Computer Science* (a.y. 2016/17, 2017/18, 2018/19, 2019/20)
- Algorithms and Data Structure, *B.Sc. in Computer Science* (a.y. 2016/17, 2019/20)

Supervision activities

University of Milano-Bicocca — *Milan, Italy*

- Co-supervisor of Davide Cozzi, PhD Student in Computer Science
- Co-supervisor of 1 M.Sc. thesis in Bioinformatics *(Eastern Africa Network of Bioinformatics Training)*
- Co-supervisor of 1 M.Sc. thesis in Computer Science
- Co-supervisor of 13 B.Sc. theses in Computer Science)

Professional Activities

Editing activities

Editor of:

- Proceedings of the 25th Conference Information Technologies – Applications and Theory (ITAT 2025)

Commitee

Program commitee member of:

- ACM BCB 2026: ACM Conference on Bioinformatics, Computational Biology, and Health Informatics
- WBCB 2025: Workshop on Bioinformatics and Computational Biology
- AIME 2024: International Conference Artificial Intelligence in Medicine
- WBCB 2023: Workshop on Bioinformatics and Computational Biology
- IEEE BIBM 2023: International Conference on Bioinformatics and Biomedicine

Organizing commitee member of:

- PhD School "Evolution in Computational Pangenomics" 2024
  Lake Como School of Advanced Studies. Como, Italy
- PhD School "Introduction to Pangenomics" 2022
  Lake Como School of Advanced Studies. Como, Italy

Reviewer Service

Journals:

- Algorithms for Molecular Biology / Bioinformatics / Bioinformatics Advances / BMC Bioinformatics / F1000Research / Genome Biology / Genome Research / Nature Communications / Nucleic Acids Research / Plant Methods / PLOS Computational Biology

Conferences:

- AIME 2024 / APBC 2022 / BIBM 2023,2022,2018 / BICOB 2018 / ICTCS 2024,2022,2017 / ISBRA 2025,2021,2020,2018 / ISMB 2026,2025 / ITAT 2023,2021 / RECOMB 2026,2025,2023,2021 / RECOMB-CG 2022 / RECOMBSEQ 2022,2021 / SPIRE 2017 / WABI 2022,2020,2019

Organizational activities

School/Workshop/Hackathon:

- Hackathon organizer at PANGAIA PhD School 2024. Como, Italy
- Hackathon organizer at ALPACA/PANGAIA Annual Workshop 2023. Amsterdam, The Netherlands
- Research group leader at ALPACA/PANGAIA Winter Wet Lab 2023. Bratislava, Slovakia

Selected Publications

For a complete list, please refer to my personal web page, my google scholar profile, and my scopus profile.

- † L. Denti, T. Krannich, T. Vinar, R. Chikhi, P. Bonizzoni, B. Brejova, F. Hormozdiari. Anyone can be the best: Impact of diverse methodologies on the evaluation of structural variant callers. bioRxiv (2025)
- † L. Denti, P. Bonizzoni, B. Brejova, R. Chikhi, T. Krannich, T. Vinar, F. Hormozdiari. Pangenome graph augmentation from unassembled long reads. bioRxiv (2025)
- S. Ciccolella, D. Cozzi, G. Della Vedova, S. Kuria, P. Bonizzoni, L. Denti. Differential quantification of alternative splicing events on spliced pangenome graphs. PLOS Computational Biology (2024)
- L. Denti*, P. Khorsand*, P. Bonizzoni, F. Hormozdiari, R. Chikhi. SVDSS: structural variation discovery in hard-to-call genomic regions using sample-specific strings from accurate long reads. Nature Methods (2023)
- L. Denti*, Y. Pirola*, M. Previtali*, T. Ceccato, G. Della Vedova, R. Rizzi, P. Bonizzoni. Shark: fishing relevant reads in an RNA-Seq sample. Bioinformatics (2020)
- L. Denti, R. Rizzi, S. Beretta, G. Della Vedova, M. Previtali, P. Bonizzoni. ASGAL: aligning RNA-Seq data to a splicing graph to detect novel alternative splicing events. BMC Bioinformatics (2018)

*Joint first/last authors are indicated with *. Any preprint is marked with a †.*

References

Prof. Paola Bonizzoni, University of Milano - Bicocca paola.bonizzoni@unimib.it
Dr. Rayan Chikhi, Institut Pasteur rayan.chikhi@pasteur.fr
Prof. Fereydoun Hormozdiari, UC Davis fhormozd@health.ucdavis.edu